Assignment 6: Implementation of Data Integrity Constraints

ProjectRapido – Bike & Cab Booking System
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ProgrammeBCA Semester 3 (2025)
CourseDatabase Management Systems
File nameAssignment_6_Data_Integrity_Constraints.pdf

1. Objective

Implement primary keys, foreign keys, UNIQUE, CHECK and DEFAULT constraints on the Rapido schema, explain each constraint, and execute SQL that proves both rejection of bad data and acceptance of valid data.

2. Primary keys

Table	Primary key	Purpose
users	user_id	Identifies each rider account
vehicles	vehicle_id	Identifies each bike/auto/cab
captains	captain_id	Identifies each driver/captain
rides	ride_id	Identifies each booking
payments	payment_id	Identifies each payment attempt/record
zones	zone_id	Identifies each service locality

captain_zones	(captain_id, zone_id)	Composite key: one membership row per pair; prevents duplicate “operates in” facts

3. Foreign keys

Foreign keys enforce the ER relationships. Captains cannot reference a missing vehicle; rides cannot reference a missing user or captain; payments cannot reference a missing ride; captain_zones cannot reference a missing captain or zone.

Declared foreign keys (from the live database):

captains

id	seq	table	from	to	on_update	on_delete	match
0	0	vehicles	vehicle_id	vehicle_id	CASCADE	RESTRICT	NONE

rides

id	seq	table	from	to	on_update	on_delete	match
0	0	captains	captain_id	captain_id	CASCADE	RESTRICT	NONE
1	0	users	user_id	user_id	CASCADE	RESTRICT	NONE

payments

id	seq	table	from	to	on_update	on_delete	match
0	0	rides	ride_id	ride_id	CASCADE	RESTRICT	NONE

captain_zones

id	seq	table	from	to	on_update	on_delete	match
0	0	zones	zone_id	zone_id	CASCADE	CASCADE	NONE
1	0	captains	captain_id	captain_id	CASCADE	CASCADE	NONE

4. UNIQUE constraints

Constraint	Purpose
users.phone_number UNIQUE	One login/OTP number maps to one rider
users.email UNIQUE	Stops two accounts sharing an email
captains.phone_number UNIQUE	One captain contact number
vehicles.registration_number UNIQUE	A registration plate belongs to one vehicle row
payments.ride_id UNIQUE	Implements 1:1 ride–payment
UNIQUE (zone_name, city)	Same locality is not inserted twice in one city

5. CHECK constraints

CHECK	Purpose
vehicle_type IN ('Bike','Auto','Cab')	Only Rapido service types
capacity BETWEEN 1 AND 6	No zero-seat or bus-sized vehicles

rating BETWEEN 0.00 AND 5.00	Star rating domain
fare >= 0 AND amount >= 0	No negative money
status IN ('Requested','Ongoing','Completed','Cancelled')	Controlled ride life-cycle
payment_method IN ('UPI','Card','Cash','Wallet')	Allowed tender types
payment_status IN ('Pending','Success','Failed','Refunded')	Allowed settlement states

6. DEFAULT constraints

Column	DEFAULT	Purpose
users.created_at / rides.requested_at	CURRENT_TIMESTAMP	Stamp row creation if the app omits a time
vehicles.capacity	1	Bike is the common Rapido case
captains.rating	0.00	New captain starts unrated
rides.status	'Requested'	A new booking is not already completed
payments.payment_status	'Pending'	Payment starts unsettled

7. SQL tests that prove integrity

Sample operational data from Assignment 7 was loaded first. Then invalid statements were executed. Each failed as required.

Test A — UNIQUE on rider phone

```
INSERT INTO users (full_name, phone_number, email) VALUES ('Fake User', '9876500001',
'fake@x.com');
```

Rejected: UNIQUE constraint failed: users.phone_number

9876500001 already belongs to Arjun Mehta, so a second rider cannot reuse it.

Test B — FOREIGN KEY on rides.user_id

```
INSERT INTO rides (user_id, captain_id, pickup_location, drop_location, fare, status) VALUES
(99, 1, 'A', 'B', 50, 'Requested');
```

Rejected: FOREIGN KEY constraint failed

User 99 does not exist, so the ride cannot be stored. This preserves the USER–REQUESTS–RIDE relationship.

Test C — CHECK on vehicle_type

```
INSERT INTO vehicles (vehicle_type, registration_number, model, capacity) VALUES ('Truck',
'KA99ZZ9999', 'X', 2);
```

Rejected: CHECK constraint failed: vehicle_type IN ('Bike', 'Auto', 'Cab')

Test D — CHECK on captain rating

```
INSERT INTO captains (full_name, phone_number, vehicle_id, rating) VALUES ('Temp',
'9000000099', 1, 6.5);
```

Rejected: CHECK constraint failed: rating BETWEEN 0.00 AND 5.00

Test E — UNIQUE 1:1 payment per ride

```
INSERT INTO payments (ride_id, amount, payment_method, payment_status) VALUES (1, 85, 'UPI',
'Success');
```

Rejected: UNIQUE constraint failed: payments.ride_id

Ride 1 already has a payment. A second row would duplicate the 1:1 relationship.

Test F — valid insert still works

```
INSERT INTO users (full_name, phone_number, email) VALUES ('Nisha Verma', '9876500099',
'nisha.verma@gmail.com');
```

Statement executed successfully.

user_id	full_name	phone_number	email
9	Nisha Verma	9876500099	nisha.verma@gmail.com

A new phone and email are accepted. Integrity rules block bad data without blocking legitimate customers.